Higher National School of Advanced Technologies (ENSTA)

SS/IA: First Semester - 2026

Module: WAMD

TD -TP 2: Create a React Application with Redux for State Management

1. Lab Objectives

At the end of this lab, students should be able to:

- Create and run a **React.js application**.
- Understand the basic structure of a React application.
- Build reusable **React components**.
- Use **JSX** to describe user interfaces.
- Manage component data using **props**.
- Manage local state using **React state**.
- Understand the role of **global state management**.
- Configure and use **Redux** in a React application.
- Create a Redux **store**.
- Define **actions** and **reducers**.
- Access and update global state from React components.
- Use **React-Redux** to connect components to the Redux store.
- Organize a React project into appropriate files and components.
- Test and debug a React application using browser Developer Tools.

2. Tools and Technologies

Tool / Technology	Purpose
Visual Studio Code	Development environment
React.js	Building the user interface
JavaScript (ES6+)	Application logic
Redux	Global state management
React-Redux	Connecting React components with Redux
Node.js / npm	Installing dependencies and running the application
Google Chrome / Firefox	Testing and debugging
Browser DevTools	Inspecting and debugging the application
Git / GitHub	Version control and project submission

3. Lab Statement

Problem

You are asked to develop a **small e-commerce web application using React.js**. The application represents an online store where users can browse products and manage a simple shopping cart.

The main objective is to move from a traditional JavaScript application to a **component-based React application** and understand why **global state management** is useful when several components need to access or modify the same data.

The application must demonstrate the use of:

- React components
- JSX
- Props
- State
- Events
- Redux Store
- Actions
- Reducers
- React-Redux

The application should provide a **clear, responsive and user-friendly interface**.

Important: This lab focuses on the React frontend and Redux. A backend, database, authentication system and payment system are not required. They will be introduced in later labs.

Proposed by: Dr. S. SENHADJI

4. Functional Requirements

The application must:

React

- Be developed using **React.js**.
- Use JSX.
- Contain multiple reusable components.
- Use props.
- Use local state where appropriate.
- Handle user events.

E-Commerce

- Display a product catalog.
- Display product information.
- Allow users to interact with products.
- Provide a shopping cart.
- Allow products to be added to the cart.
- Allow products to be removed from the cart.
- Dynamically update the interface when the cart changes.

Redux

- Use a Redux store.
- Define actions.
- Define reducer logic.
- Use React-Redux.
- Read global state from React components.
- Dispatch actions from React components.
- Display the updated global state.

Interface

- Provide a clear and user-friendly interface.

5. Optional Features

Students who complete the required functionality may implement additional features.
Examples include:

- Product search
- Category filtering
- Product details
- Quantity management
- Cart total
- Favorites
- Product sorting
- Dark/light mode
- Simple navigation
- Local storage
- Discount display

These features are **optional**.
The priority is to correctly implement **React, components, state and Redux**.

6. Creative Requirement

The **application theme and visual identity are intentionally left open to creativity**.
Students are free to choose:

- Store name
- Logo
- Product category

- Products
- Colors
- Typography
- Layout
- Images
- Navigation style
- Product-card design
- Cart design
- Additional features

For example:
DzTech Store → Electronics
DzFashion → Clothing
BookDZ → Books
SportZone → Sports
ArtisanDZ → Handmade products
Students may propose another appropriate e-commerce concept.
Creativity is encouraged, but all required technical functionalities must be respected.

7. Deliverables

Students should submit their work through the **course Classroom**.
The submission must contain:

1. **GitHub repository link** containing the complete React project.
2. **Screenshots of the final application**.
3. A README.md containing:
    - Student name
    - Application/store name
    - Application description
    - Technologies used
    - Main React components
    - Redux state managed by the application
    - Implemented features
    - Instructions for running the application

Optional: Students may deploy their React application online and provide the live URL. Short demonstration video.